

BioKB Integration and Knowledge Graph Construction to explore mediterranean plants usage in wound healing

Programming Lab 2 – Group 3

Mehak Nadeem

Özlem Kalkan

Parmida Zamani

Rojin Sadat Yasini

Introduction

The goal of this project is to generate a pipeline to generate a comprehensive knowledge graph to explore potential compounds in mediterranean plants that treats wounds on skin.

Databases

- Plant taxonomy → IPNI, WFO, WCVP, taxtree
- Chemicals → ChEBI, COCONUT
- Enzymes → BRENDA
- Literature → Ethnobotv2
- Literature and plant species → Ethnobotv1

Methodology

We implemented:

1. Automated BioKB import
2. RDF/Turtle export of BioKB
3. Import BioKB to Neo4j
4. Ethnobotv1, Ethnobotv2 import
5. Turtle export of Ethnobotv1, Ethnobotv2
6. Import Ethnobotv1, Ethnobotv2 to Neo4j
7. Neo4j graph construction
8. Cross-domain Cypher queries

Automated BioKB import

- python main.py

```
engine = create_engine(  
    "mysql+pymysql://group_project_user:group_project_password@localhost:3306/group_project_db"  
)  
  
dbm = DbManager(engine)  
  
dbm.import_biokb_dbs()
```

Turtle export of BioKB

```
dbm.create_turtle_files()
```

example:

```
dbm.create_chebi_turtle_files()  
dbm.create_brenda_turtle_files()
```

Import BioKB to Neo4j

```
dbm.import_neo4j(  
    neo4j_uri="bolt://localhost:7687",  
    neo4j_user="neo4j",  
    neo4j_password="neo4j_password",  
)
```

Ethnobotv2 scripts

```
#import generate_turtle_file and load_to_neo4jscripts first.
from plantgraph3.ethnobot_v2 import generate_turtle_file as s2t
from plantgraph3.ethnobot_v2 import load_to_neo4j as loader
import sqlite3
```

```
6
7 import os
8 from sqlalchemy import create_engine, text
9 from rdflib import Graph, Namespace, Literal, URIRef
10 from rdflib.namespace import RDF, RDFS, XSD
11
12 SQLITE_URL = "sqlite:///ethnobot_v1.db"
13 OUTPUT_DIR = "import"
14
15 # Define Namespaces
16 ETHNOBOT = Namespace("http://ethnobot.org/")
17 EB_CLASS = Namespace("http://ethnobot.org/class/")
18 EB_PROP = Namespace("http://ethnobot.org/property/")
19
20 os.makedirs(OUTPUT_DIR, exist_ok=True)
21
22
23 def new_graph():
24     g = Graph()
25     g.bind("ethnobot", ETHNOBOT)
26     g.bind("eb", EB_CLASS)
27     g.bind("ebp", EB_PROP)
28     g.bind("xsd", XSD)
29     g.bind("rdfs", RDFS)
30     return g
31
32
33 def safe_literal(value, datatype=None):
34     if value is None:
35         return None
36     if datatype:
37         return Literal(value, datatype=datatype)
38     return Literal(str(value))
39
40
41 def save_graph(g, filename):
42     path = os.path.join(OUTPUT_DIR, filename)
43     g.serialize(destination=path, format="turtle")
44     print(f" Saved: {path} ({len(g)} triple)")
```

Turtle generation of Ethnobotv2

```
from pathlib import Path
from sqlalchemy import create_engine

conn = sqlite3.connect("prog_lab2.db")
cursor = conn.cursor()

#Create Turtle files for ethnobotanical data prog_lab2.db

s2t.export_journals(cursor)
s2t.export_articles(cursor)
s2t.export_tables(cursor)
s2t.export_disease_taxon(cursor)

conn.close()
```

Turtle generation of Ethnobotv1

```
from plantgraph3.ethnobot_v1.clean_ethnobot_db import EthnobotDBCleaner
```

- before generating turtle file, redundant tables and empty tables are removed

```
from plantgraph3.ethnobot_v1 import generate_turtle_v1 as gen
```

- Table left after removal:
conservation_status, region_type, specimen_part, usage_type,
pharmacological_effect, molecule, mode_of_preparation, mode_of_administration,
reference, region, species, traditional_usage, formulation,
species_region_association, formulation_mode_of_administration_association,
plant_specimen, plant_specimen_specimen_part_association, eb_index, ethnic_info

```
SQLITE_URL = "sqlite:///ethnobot_v1.db"
OUTPUT_DIR = "import"
```

Python

```
#remove redundant tables before importing database
TABLES_TO_DROP = [
    "plant_specimen_pharmacological_effect_association",
    "plant_specimen_mode_of_preparation_association",
    "plant_specimen_molecule_association",
    "plant_specimen_toxicological_effect_association",
    "plant_specimen_traditional_usage_association",
    "formulation_adverse_effect_association",
    "formulation_molecule_association",
    "formulation_region_association",
    "species_habitat_association",
    "adverse_effect",
    "toxicological_effect",
    "habitat",
]

cleaner = EthnobotDBCleaner("ethnobot_v1.db")
cleaner.drop_tables(TABLES_TO_DROP)
cleaner.list_tables()
```

Import Ethnobotv1 to Neo4j

```
from plantgraph3.ethnobot_v1 import load_to_neo4j_v1 as loader
```

Neo4j graph construction

The knowledge graph integrates multiple biological domains:

- Plants (taxonomy databases)
- Chemical compounds (ChEBI, COCONUT)
- Enzymes (BRENDA)
- Ethnobotanical literature

Relationships connect entities across databases:

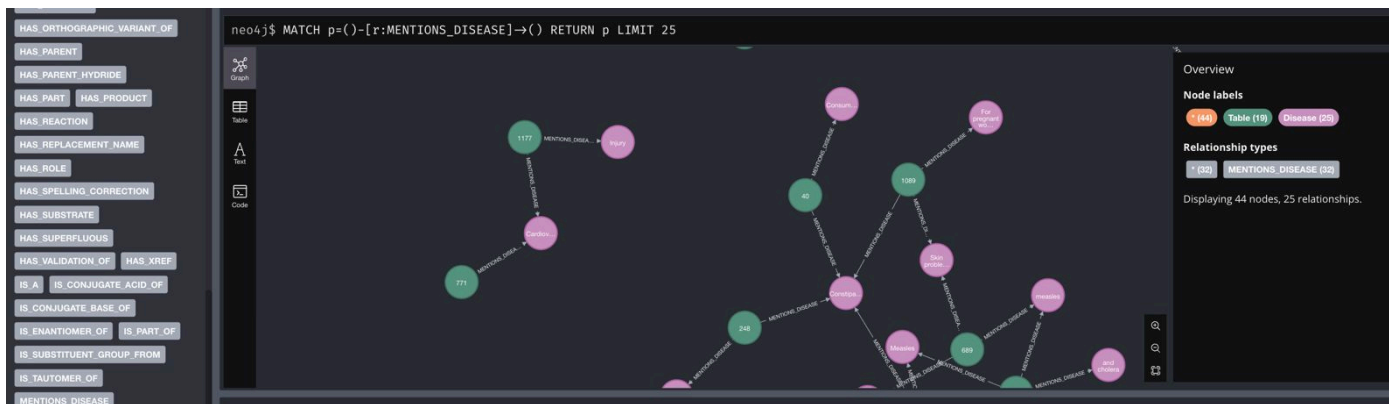
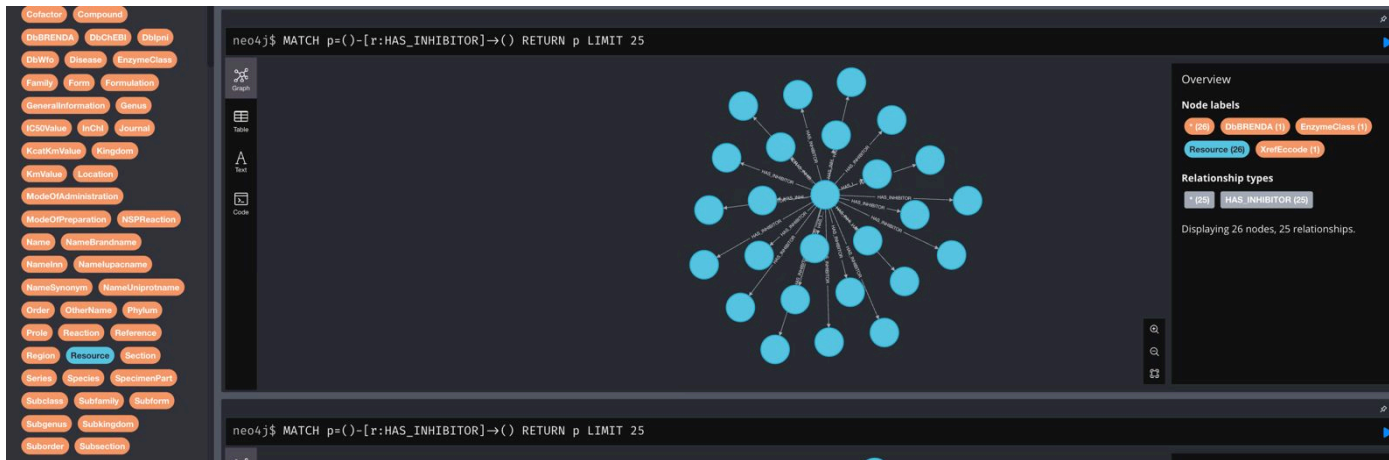
Plant → produces → Compound

Compound → interacts with → Enzyme

Article → mentions → Plant

Impact: Enables cross-database biological discovery

Neo4j graph construction



Neo4j is accessed via:

<http://localhost:7474>

Why Graph Database Instead of Relational Database?

Biological data is highly connected.

Relational database requires complex JOINS:

Plant → Compound → Enzyme → Disease

Graph database advantages:

- Direct relationship traversal
- Faster multi-step queries
- Natural representation of biological networks
- Scalable for millions of nodes

Automated Data Integration Pipeline

Pipeline steps:

BioKB databases



Turtle file generation



Neo4j graph import



Graph construction



Cypher queries and analysis

Fully automated integration pipeline.

Reproducibility and Deployment

Ensured through:

- Containerized environment (Docker/Podman)
- Version-controlled source code
- Automated database import pipeline

Benefits:

- Easy replication of results
- Consistent environment
- Supports collaborative research

Modular System Architecture

Library Design

Project components:

- Docker containers – isolate services
- Neo4j database – graph storage
- MySQL database – BioKB storage
- Python scripts – automation and import
- Configuration files – environment setup

Modular System Architecture

```
pyproject.toml
1 [project]
2   name = "plantgraph3"
3   version = "0.1.0"
4   description = "Add your description here"
5   readme = "README.md"
6   requires-python = ">=3.12"
7   dependencies = [
8     "biokb_ipni @ git+https://github.com/biokb/biokb_ipni.git",
9     "biokb_taxtree @ git+https://github.com/biokb/biokb_taxtree.git",
10    "biokb_wcvp @ git+https://github.com/biokb/biokb_wcvp.git",
11    "biokb_wfo @ git+https://github.com/biokb/biokb_wfo.git",
12    "biokb-brenda @ git+https://github.com/biokb/biokb_brenda.git",
13    "biokb_coconut @ git+https://github.com/biokb/biokb_coconut.git",
14    "biokb_chebi @ git+https://github.com/biokb/biokb_chebi.git",
15    "sqlalchemy>=2.0.46",
16    "pandas>=3.0.0",
17    "neo4j>=6.1.0",
18  ]
19
20 [build-system]
21 requires = ["uv_build>=0.10.0,<0.11.0"]
22 build-backend = "uv_build"
23
24 [dependency-groups]
25 dev = [
26   "ipykernel>=7.2.0",
27 ]
28 docs = [
29   "mkdocs>=1.6.1",
30   "mkdocs-jupyter>=0.24.1",
31   "mkdocs-material>=9.7.1",
32   "mkdocstrings[python]>=1.0.3",
33 ]
34 tests = [
35   "pytest>=9.0.2",
36   "pytest-cov>=7.0.0",
37 ]
```

Workflow

1. Return article with plants and diseases that plants have a role
2. Filter by skin wound types
3. Search mediterranean countries contained in articles names and filter
as a result we obtained Mediterranean plants involving in skin wound types
diseases
4. We returned the compounds from Chebi database that playing a role in wound
processes

Cypher queries

```
WITH
[ {Wound Terms} ] AS woundList,

[ {Mediterranean Countries} ] AS medCountries

MATCH (a:Article)<-[:PART_OF]-(tb:Table)
      -[:MENTIONS_TAXON]->(t:Taxon)
      -[:USED_FOR]->(d:Disease)

WHERE
  ANY(w IN woundList
    WHERE toLower(coalesce(d.diseaseName,d.name,'')) = toLower(w))
  AND ANY(country IN medCountries
    WHERE toLower(a.title) CONTAINS country)

RETURN
  a.title AS article,
  t.taxonName AS plant,
  t.ipni AS ipni,
  coalesce(d.diseaseName, d.name) AS disease
LIMIT 200;
```

Cypher queries

article	plant	ipni	disease
Traditional Medicine in the Pristi	Plantago lanceolata	321285-2	Wound
Traditional Medicine in the Pristi	Arctium lappa	178385-1	Wound healing
Traditional Medicine in the Pristi	Arctium lappa	178385-1	wounds
Traditional Medicine in the Pristi	Arctium lappa	178385-1	Skin wounds
Traditional Medicine in the Pristi	Hypericum perforatum	433719-1	Wounds
Traditional Medicine in the Pristi	Hypericum perforatum	433719-1	Wound
Traditional Medicine in the Pristi	Rubus fruticosus	736933-1	skin wounds
Traditional Medicine in the Pristi	Urtica dioica	260630-2	skin wounds
Traditional Medicine in the Pristi	Salvia officinalis	456833-1	Wound
Resilience at the border: traditio	Solanum tuberosum	null	burns
Resilience at the border: traditio	Achillea millefolium	2294-2	Skin lesions
Resilience at the border: traditio	Achillea millefolium	2294-2	Wound healing
Resilience at the border: traditio	Nicotiana tabacum	null	Wound
Resilience at the border: traditio	Plantago lanceolata	321285-2	Wound
Resilience at the border: traditio	Plantago major	321286-2	Skin wounds

Cypher queries

```
query =""" MATCH (c:DbChEBI)
WHERE toLower(coalesce(c.definition, '')) CONTAINS "wound"
RETURN c.ascii_name, c.definition
LIMIT 50"""
pd.DataFrame(query_neo4j(query))
```

Cypher queries

```
neo4j$ MATCH (c:DbCHEBI) WHERE toLower(coalesce(c.definition,'')) CONTAINS "wound" RETURN c.ascii_name, c.definition LIMIT 50
```

	c.ascii_name	c.definition
1	"nitrofurazone"	"A semicarbazone resulting from the formal condensation of semicarbazide with 5-nitrofuraldehyde. A broad spectrum antibacterial drug, although with little activity against <i>Pseudomonas</i> species, it is used
2	"piperonylic acid"	"A member of the class of benzodioxoles that is 1,3-benzodioxole substituted by a carboxy group at position 5. It is a natural product isolated from several plant species. It is a selective mechanism-based inactivator of
3	"grandiflorenic acid"	"A tetracyclic diterpenoid with formula C ₂₀ H ₂₈ O ₂ which exhibits anti-inflammatory, antibacterial, antileishmanial and wound-healing properties.
4	"traumatic acid"	"A monounsaturated straight-chain dicarboxylic acid with the double bond at C-2; a plant wound-healing hormone."
5	"asiaticoside"	"A triterpenoid saponin that is a trisaccharide derivative of asiatic acid. Isolated from <i>Centella asiatica</i> , it exhibits anti-inflammatory, antioxidant and wound-healing properties."
6	"3,6-diaminoacridine"	"An aminoacridine that is acridine that is substituted by amino groups at positions 3 and 6. A slow-acting bacteriostat that is effective against many Gram-positive bacteria (but ineffective against spores), its salts were

Cypher queries

```
MATCH (t:DbNCBITaxTree:Taxon)
      <-[:SAME_AS]-(c:DbCoconut)
RETURN t.scientific_name, c.name
LIMIT 50
```

t.scientific_name	c.name
"Astrogorgia sp."	"Astrogorgia sp."
"Pittosporum angustifolium"	"Pittosporum angustifolium"
"Polygala amara"	"Polygala amara"
"Cassipourea guianensis"	"Cassipourea guianensis"
"Amanita cokeri"	"Amanita cokeri"
"Asteriscus graveolens"	"Asteriscus graveolens"
"Baccharis megapotamica"	"Baccharis megapotamica"
"Bombus muscorum"	"Bombus muscorum"
"Centrosema pascuorum"	"Centrosema pascuorum"
"Ceratozamia fuscoviridis"	"Ceratozamia fuscoviridis"

Contribution – Issue Board

in total 30 issues

☐ upload and generate presentation markdown file bit/plab2/25/group_projects/plantgraph3#30 · created 2 hours ago by Özlem Kalkan	Closed  closed 45 seconds ago
☐ testing cypher queries bit/plab2/25/group_projects/plantgraph3#29 · created 2 hours ago by Özlem Kalkan	Closed  closed 41 seconds ago
☐ create wound healing related cypher queries bit/plab2/25/group_projects/plantgraph3#28 · created 2 hours ago by Özlem Kalkan	Closed   2 closed 1 minute ago
☐ create test queries bit/plab2/25/group_projects/plantgraph3#27 · created 1 day ago by Rojinsadat Yasini	Closed  closed 7 hours ago
☐ Create a jupyter notebook to clean the ethnobot_v1 db bit/plab2/25/group_projects/plantgraph3#26 · created 1 day ago by Özlem Kalkan	Closed   1 closed 1 day ago
☐ created a folder for ethnobot_v2 (prog_lab2) database , prepare a python script bit/plab2/25/group_projects/plantgraph3#25 · created 1 day ago by Özlem Kalkan	Closed   2 closed 7 hours ago
☐ create namespaces.py file bit/plab2/25/group_projects/plantgraph3#24 · created 1 day ago by Rojinsadat Yasini	Closed   1 closed 1 day ago
☐ adding drug database bit/plab2/25/group_projects/plantgraph3#23 · created 2 days ago by Özlem Kalkan	Closed   1 closed 1 day ago
☐ what is demo notebook? should we create a demo file ? bit/plab2/25/group_projects/plantgraph3#22 · created 2 days ago by Özlem Kalkan	Closed   2 closed 2 days ago
☐ Write the Project Reprt bit/plab2/25/group_projects/plantgraph3#21 · created 2 days ago by Parmida Zamani	Closed   1 closed 1 minute ago
☐ Integrate BioKB import pipeline into unified main.py script bit/plab2/25/group_projects/plantgraph3#20 · created 3 days ago by Mahak Nadeem	Closed   1 closed 3 days ago
☐ Verify Neo4j import and analyze graph schema	Closed 

Contribution – Commits

bit / plab2 / 25 / group_projects / plantgraph3

P

plantgraph3

🔔

☆ Star 0

🍴 Fork 0

⋮

🔗 main

+ 🔍 Find file

Code

⋮

 Merge branch 'dev' into 'main' ⋮
Özlem Kalkan authored 32 minutes ago

daf0284c

History

Name	Last commit	Last update
📁 notebooks	Merge branch 'dev' into 'main'	32 minutes ago
📁 src	deleted duplicated and unwanted files	3 hours ago
📄 .DS_Store	last changes	6 hours ago
📄 .env	correct structure and organization plus ...	1 day ago
📄 .gitignore	correct structure and organization plus ...	1 day ago
📄 .python-version	initial db	1 month ago
📄 Dockerfile	example project was created	1 week ago
📄 Project_Report.md	Corrected text	21 hours ago
📄 README.md	Edits in README	3 hours ago
📄 docker-compose.yml	example project was created	1 week ago
📄 main.py	Edits in README	3 hours ago
📄 pyproject.toml	example project was created	1 week ago

Project information

🔗 63 Commits

🔗 3 Branches

🏷️ 0 Tags

💾 119.5 MiB Project Storage

📄 README

+ Add LICENSE

+ Add CHANGELOG

+ Add CONTRIBUTING

+ Enable Auto DevOps

+ Add Kubernetes cluster

+ Set up CI/CD

+ Add Wiki

+ Configure Integrations

Created on

January 08, 2026

Contribution – Commits

 **wearethechampion** 

  Star 0  Fork 0 

 main 

  Find file **Code**  

 **Upload New File**
Özlem Kalkan authored 1 month ago

0c7103e3



History

Name	Last commit	Last update
 go_terms	Upload New File	1 month ago
 sqlite_database	sqlite db	2 months ago
 .python-version	cleaning table	2 months ago
 README.md	Initial commit	2 months ago
 main.py	cleaning table	2 months ago
 pyproject.toml	cleaning table	2 months ago
 uv.lock	cleaning table	2 months ago

Project information

 16 Commits

 2 Branches

 0 Tags

 385.5 MiB Project Storage

 README

 [Add LICENSE](#)

 [Add CHANGELOG](#)

 [Add CONTRIBUTING](#)

 [Enable Auto DevOps](#)

 [Add Kubernetes cluster](#)

 [Set up CI/CD](#)

 [Add Wiki](#)

Thank You

Questions?